

Engineering Physics: BAPHY105

Particle in a Box

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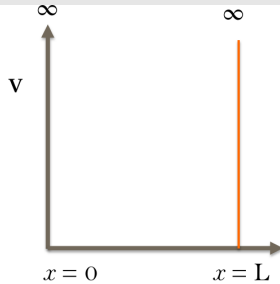
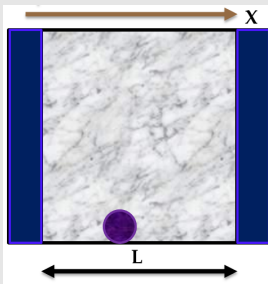
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Lecture Outcomes

- 1 Particle confined inside an 1-D potential Region
 - Solution to the Schrödinger equation
 - Boundary conditions
 - Normalization of the wavefunction
 - Graphical representation

Particle in an 1-D box: Problem Definition

Consider a particle of mass m confined in an 1-dimensional potential region of length L defined along the x -axis.



- ✓ The potential can be expressed in the form,

$$V(x) = \begin{cases} \infty & \text{if } x \leq 0 \text{ or } x \geq L \\ 0 & \text{if } 0 < x < L \end{cases}$$

- ✓ Wavefunction corresponding to the particle must **vanish at infinite potential**, i.e.

$$\Psi(x) = 0 \text{ for } x \leq 0 \text{ and } x \geq L$$

- ✓ One dimensional Schrödinger's equation for this problem,

$$\frac{d^2\Psi(x)}{dx^2} + \frac{2mE}{\hbar^2} (E - V(x)) \Psi(x) = 0$$

- ✓ Inside the potential box, **$V = 0$** .

$$\Rightarrow \frac{d^2\Psi(x)}{dx^2} + \frac{2mE}{\hbar^2} E \Psi(x) = 0$$

- ✓ In the absence of any potential inside the box, **energy of the particle is completely kinetic.**

$$\frac{d^2\Psi(x)}{dx^2} + \frac{2mE}{\hbar^2}\Psi(x) = 0$$

- ✓ Substituting $k = \sqrt{\frac{2mE}{\hbar^2}}$, (known as the wave number)

$$\frac{d^2\Psi(x)}{dx^2} + k^2\Psi(x) = 0$$

- ✓ **Auxillary equation** for the above equation:

$$D^2 + k^2 = 0, \quad \Rightarrow D = \pm ik$$

- ✓ Complete solution to the Schrödinger equation is,

$$\Psi(x) = Ae^{ikx} + Be^{-ikx}$$

✓ Boundary conditions are used to evaluate the unknown coefficients A and B .

✓ First boundary condition, $\Psi(x) = 0$ at $x = 0$.

$$\Rightarrow 0 = A \cdot 1 + B \cdot 1 \quad \Rightarrow B = -A$$

$$\Rightarrow \Psi(x) = A \left(e^{ikx} - e^{-ikx} \right) = -2iA \sin(kx) = C \sin(kx)$$

$$\therefore \Psi(x) = C \sin(kx)$$

✓ Second boundary condition, $\Psi(x) = 0$ at $x = L$.

$$\Rightarrow 0 = C \sin(kL) \quad \Rightarrow kL = n\pi \quad (n = 1, 2, 3, \dots)$$

$$\Rightarrow k = \frac{n\pi}{L} \quad \Rightarrow E_n = \frac{n^2 \pi^2 \hbar^2}{2mL^2}$$

✓ Wavenumber k as well as the energy eigenvalue E_n are quantized.

- Eigenfunction $\Psi_n(x)$ and the eigenvalue E_n are:

$$\Psi_n(x) = C \sin\left(\frac{n\pi x}{L}\right), E_n = \frac{n^2 \pi^2 \hbar^2}{2mL^2}$$

- ✓ Since the probability of finding the particle inside the region $0 < x < L$ is unity,

$$\int_0^L \Psi_n^*(x) \Psi_n(x) dx = 1$$

$$\Rightarrow |C|^2 \int_0^L \sin^2\left(\frac{n\pi x}{L}\right) dx = 1$$

$$\Rightarrow C^2 \left\{ \int_0^L \left[\frac{1 - \cos\left(\frac{2\pi nx}{L}\right)}{2} \right] dx \right\} = 1$$

$$\Rightarrow \frac{C^2}{2} \left[x - \frac{L}{2\pi n} \sin \left(\frac{2\pi n x}{L} \right) \right] \Big|_0^L = 1$$

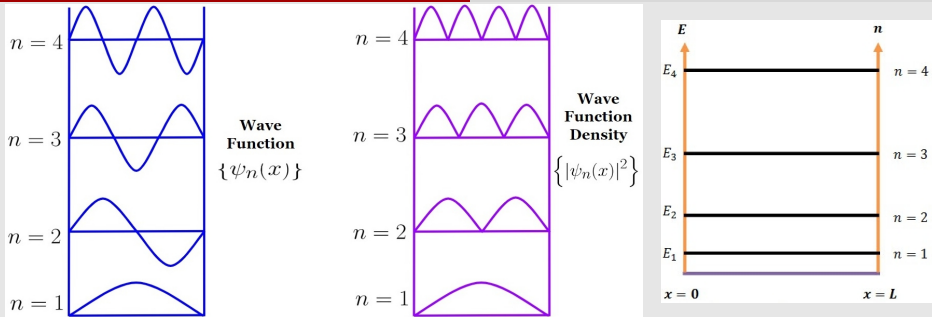
$$\Rightarrow C = \sqrt{\frac{2}{L}}$$

- Normalized wavefunction (eigen function):

$$\Psi_n(x) = \sqrt{\frac{2}{L}} \sin \left(\frac{n\pi x}{L} \right)$$

- Energy is quantized and the **ground state energy is non-zero**.

$$E_1 = \frac{\pi^2 \hbar^2}{2mL^2}$$



- The same idea can be extended to 3-D box (cube of side a):

$$\Psi_n = \sqrt{\frac{8}{a^3}} \sin\left(\frac{n_x \pi x}{a}\right) \sin\left(\frac{n_y \pi y}{a}\right) \sin\left(\frac{n_z \pi z}{a}\right)$$

$$E_n = \frac{\pi^2 \hbar^2}{2ma^2} (n_x^2 + n_y^2 + n_z^2)$$

- n : QUANTUM NUMBER